



Generative AI usage and sustainable supply chain performance: A practice-based view

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ABSTRACT

The emergence of generative AI presents numerous potential solutions to address challenges in sustainable supply chain management (SCM). However, not all firms can effectively master the methods of using generative AI and realize potential benefits. To address this dilemma, we adopt a practice-based view (PBV) to examine generative AI usage's effect on sustainable supply chain performance (SSCP). Analyzing survey data from 213 Chinese manufacturing firms, we identify a positive relationship between generative AI usage and SSCP. Moreover, two types of sustainable supply chain practices—green supply chain collaboration (GSCC) and circular economy implementation (CEI)—emerge as serial mediators connecting this relationship. We contribute to existing AI-enabled SCM research by elucidating the potential mediation mechanisms underlying the link between generative AI usage and SSCP. We also offer insightful implications for firms adapting to new norms in global SCM.

1. Introduction

Firms often face numerous challenges in achieving supply chain sustainability, such as managing resource efficiency, reducing carbon footprints, ensuring ethical sourcing, and maintaining resilience against disruptions (Xu et al., 2022b). These challenges highlight the need for novel solutions to address both operational disruptions and sustainability goals (Kargar et al., 2024). Fortunately, generative AI's ability to create innovative and personalized solutions potentially addresses these challenges (Fosso Wamba et al., 2023). However, differences in technological adoption, organizational readiness, and specific supply chain contexts mean the use of generative AI in supply chain management (SCM) is inconsistent (IBM, 2024). Moreover, although generative AI holds promise, its effective use requires firms to possess specific capabilities and understanding that are not universally present (IBM, 2024). Therefore, scholars and practitioners need to delve deeper into the mechanisms by which generative AI usage influences sustainable supply chain performance (SSCP).

Existing AI-enabled SCM literature primarily highlights AI's value in sensing, interacting, learning, and decision-making (Pournader et al., 2021), along with its roles in bolstering supply chain agility, supply chain resilience, and supply chain finance (Dubey et al., 2022; Olan et al., 2022b; Gupta et al., 2023; Dey et al., 2023). Although previous studies provide significant insights, they

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focus predominantly on discriminative AI, rather than generative AI (Li et al., 2024). Several prominent studies have discussed generative AI's potential in operations and SCM; however, most are conceptual or descriptive (Richey et al., 2023; Jackson et al., 2024), and large-scale investigations are relatively scarce (Fosso Wamba et al., 2024). Because of the heterogeneity in the use of generative AI, we aim to bridge the gap by providing concrete empirical evidence on generative AI's application and influence in sustainable SCM.

The practice-based view (PBV) suggests that practice refers to activities firms might execute, and even small day-to-day activities may lead to performance differences across firms (Bromiley and Rau, 2014). Generative AI is an important technological asset for firms, but what firms do with that asset is more important (Sirmon et al., 2011). This implies that the value of generative AI can only be maximized when firms effectively embed it into their practices. To better illustrate this process, we consider two sustainable supply chain practices: green supply chain collaboration (GSCC) and circular economy implementation (CEI). We chose these two practices because they represent distinct but complementary approaches to achieving sustainability goals. GSCC focuses on collaborative efforts among supply chain partners to achieve environmental goals (Luo et al., 2014), whereas CEI promotes resource reduction, reuse, and recycling (Kristoffersen et al., 2021). The combination of these practices allows us to explore a broader spectrum of sustainability strategies and how generative AI can be utilized to enhance both collaborative and resource-efficient efforts. While GSCC and CEI have been studied independently in the context of supply chain sustainability (Yang et al., 2013; Bag et al., 2022), there is a gap in understanding how firms can optimally embed generative AI to promote both GSCC and CEI effectively within a single framework. This gap impedes the development of best practices that could guide firms in maximizing the benefits of generative AI and achieving robust sustainability outcomes. Hence, our research question is: *How do GSCC and CEI link the generative AI usage–SSCP association?*

To address this question, we surveyed 213 Chinese manufacturing firms. The empirical results enrich the existing SCM literature in several ways. First, although existing studies have discussed the role of discriminative AI in sustainable SCM (Hendriksen, 2023), generative AI has received limited attention. Unlike discriminative AI, which improves supply chain sustainability through enhanced existing operations (Mithas et al., 2022), we provide novel insights by highlighting the distinctive role of generative AI in sustainable SCM.

Second, although several leading studies have offered conceptual or descriptive insights into generative AI's potential to support operations and SCM (Richey et al., 2023; Jackson et al., 2024), large-scale investigations are relatively scarce (Fosso Wamba et al., 2024). Drawing on the PBV, we provide concrete empirical evidence on the effect of generative AI usage on SSCP. This validates PBV's effectiveness in the context of generative AI usage and provides valuable, data-driven insights to supplement existing AI-enabled SCM research.

Third, previous studies have recognized the role of GSCC and CEI in enhancing sustainable performance individually (Luo et al., 2014; Marrucci et al., 2022). Our study integrates them into a single framework and highlights their mediating role in linking the generative AI usage–SSCP association and their potential connections. These findings enrich the current understanding of the role of GSCC and CEI by demonstrating how firms can effectively manage and utilize generative AI to enhance supply chain sustainability.

2. Literature review

2.1. Discriminative and generative AI for sustainable SCM

Sustainable SCM aims to integrate environmental, social, and economic considerations into supply chain operations (Nayal et al., 2022b). Although many emerging technologies, such as the Internet of Things (IoT) (Manavalan and Jayakrishna, 2019), blockchain (Yousefi and Tosarkani, 2022), and big data analytics (Raut et al., 2021), contribute significantly to sustainable SCM, there is a growing interest in understanding the roles of discriminative and generative AI in this context. Discriminative AI is particularly adept at identifying patterns within large datasets and making informed, data-driven decisions (Li et al., 2024). This capability is pivotal for optimizing resource use, predicting fluctuations in demand, and ultimately enhancing the overall efficiency of supply chain operations (Bhattacharya et al., 2024). By leveraging discriminative AI, firms can streamline their processes, reduce waste, and make more accurate forecasts, all of which contribute to a more sustainable supply chain (Khan et al., 2023).

However, unlike discriminative AI, which enhances existing operations (Mithas et al., 2022), generative AI introduces unique capabilities that extend beyond traditional AI applications (Dubey et al., 2024). Generative AI can rapidly generate and evaluate innovative solutions, such as eco-friendly packaging designs and energy efficiency improvements (Huawei, 2024). This feature allows firms to quickly respond to market demands and environmental changes, developing more sustainable products and services. Additionally, generative AI can create simulated data to test and optimize supply chain operations in a virtual environment, identifying potential environmental and social costs (Deloitte, 2023). For example, generative AI can simulate different logistics routes or production processes, enabling firms to choose the most sustainable option. Furthermore, generative AI can analyze vast amounts of personalized data and market trends, helping firms develop more tailored and flexible supply chain strategies (IBM, 2024).

Given these distinct characteristics, generative AI plays a pivotal role in sustainable SCM by fostering innovation and personalized solutions (Dubey et al., 2024; Fosso Wamba et al., 2024). Although several studies discuss how firms could leverage generative AI to enhance operations and SCM, many remain conceptual or descriptive (Richey et al., 2023; Jackson et al., 2024). There is a notable gap in large-scale empirical investigations that rigorously assess generative AI's practical influence and implementation challenges in sustainable SCM (Fosso Wamba et al., 2023). This gap exists partly because of the diverse adoption rates of generative AI across industries, varying levels of organizational readiness to implement generative AI, and the unique supply chain contexts within which firms operate (IBM, 2024). Hence, scholars should conduct comprehensive empirical studies evaluating generative AI's benefits, challenges, and performance effects across different supply chain environments. This would provide valuable insights into how

generative AI can effectively integrate into sustainable SCM, tailored to meet specific industry needs and operational contexts.

2.2. Sustainable supply chain practices

Sustainable supply chain practices have attracted significant attention recently because of increasing environmental concerns and regulatory pressures (Schilling and Seuring, 2024). Among these practices, two prominent approaches are GSCC and CEI (Yang et al., 2013; Bag et al., 2022). GSCC refers to the strategic collaboration between a firm and its supply chain partners to implement environmentally friendly initiatives effectively (Luo et al., 2014). This practice includes shared environmental goals, joint environmental management systems, and collaborative efforts to reduce environmental repercussions (Gunasekaran et al., 2015). Research has identified several drivers for GSCC, including regulatory pressure, market demand for environmentally friendly products, and corporate social responsibility (Xu et al., 2022a; Ardakani et al., 2023). Moreover, the effects of GSCC are multifaceted, leading to improved environmental performance, cost savings, and enhanced reputation (Yang et al., 2013).

CEI delineates how firms proficiently execute value-added activities by embracing circular strategies (Kristoffersen et al., 2021). These strategies include recycling, remanufacturing, refurbishing, and designing products for disassembly (Meier et al., 2023). CEI aims to establish a regenerative system that minimizes waste and continuously reuses resources, creating a sustainable loop (Zhang et al., 2021). The primary drivers behind CEI are multifarious. Regulatory requirements play a significant role, compelling firms to adhere to environmental standards and legislation (Bag et al., 2022). Resource scarcity is another critical driver, pushing firms to seek innovative ways to use materials more efficiently (Di Domenico et al., 2023). Finally, CEI practices have profound effects, promoting resource efficiency, reducing environmental pollution, and creating new business opportunities (Farooque et al., 2022; Dey et al., 2022; Nayal et al., 2022a).

Existing research also highlights that technologies enhance GSCC and CEI (Meier et al., 2023). Technologies such as the IoT, blockchain, and big data analytics facilitate better data sharing, transparency, and tracking of environmental metrics across the supply chain (Manavalan and Jayakrishna, 2019; Zhou et al., 2023; Bag et al., 2023), enabling real-time monitoring and more efficient supply chain operations. Although discriminative AI can help firms improve decision-making, such as green supplier selection and waste sorting (Wang et al., 2023), it primarily focuses on classification tasks by distinguishing between different categories. Contrastingly, generative AI is commonly employed in image generation, text generation, speech synthesis, and data augmentation (Fosso Wamba et al., 2023), offering new ways to model and simulate supply chain scenarios, predict outcomes, and develop sustainable strategies. Given that generative AI presents new opportunities and challenges (Kshetri et al., 2023), there is a growing need to explore its influence on sustainable supply chain practices. This exploration may help firms achieve greater efficiency and sustainability in their supply chain operations.

3. Theory and hypotheses

3.1. PBV

The PBV emerged relatively recently, introduced by Bromiley and Rau (2014) as a distinct lens diverging from the traditional resource-based view (RBV). Specifically, the RBV asserts that a firm's unique resources and capabilities are the main drivers of its competitive advantage (Wernerfelt, 1984). These can be tangible (e.g., physical assets) or intangible (e.g., brand reputation, patents). However, the RBV is often criticized for being static because it focuses on what firms have rather than how they use their resources and adapt over time (Sirmon et al., 2011). Conversely, the PBV focuses on firm performance (Dubey et al., 2022). Moreover, the PBV contends that firms exhibit variations in their embrace of seemingly simple practices, and these distinctions serve as the bedrock for performance differentials among firms (Bromiley and Rau, 2014). Thus, effective practices are the sources of performance improvement.

Although the PBV is a relatively new lens, SCM scholars have utilized it to comprehend some emerging phenomena (Malacina et al., 2022). For example, analyzing responses from 171 non-governmental organizations, Dubey et al. (2022) find that in the humanitarian supply chain context, agility and resilience are two critical factors in the relationship between AI-driven big data analytics capability and supply chain performance. Employing a fixed effects model to analyze panel data from Chinese manufacturing firms, Tian et al.

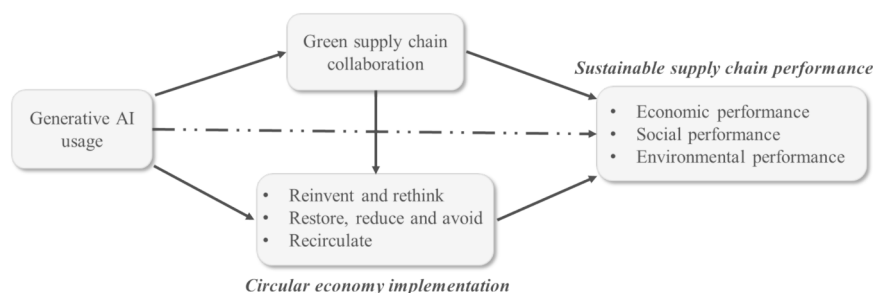


Fig. 1. Research framework.

(2023) investigate how digital transformation practices influence a firm's operational efficiency. Through a sample of 639 valid survey responses, Khan et al. (2023) confirm the positive effects of sustainable supply chain practices on sustainable performance within the textile industry context.

Building on previous research, particularly the work of Dubey et al. (2022), the PBV should help explain the mechanisms behind the generative AI usage–SSCP association. Specifically, the PBV highlights that the value derived from resources depends significantly on their effective integration into organizational practices (Bromiley and Rau, 2014). In our study, this implies that generative AI usage's benefits for SSCP are realized only when it is seamlessly incorporated into the firm's sustainable supply chain practices. Because the PBV provides a framework in which practices are intermediate constructs in the relationship between explanatory variables and the performance outcome of interest (Bromiley and Rau, 2014), we draw on the PBV and propose our conceptual model in Fig. 1. Below, we outline each hypothesis for examination.

3.2. Effects of generative AI usage on SSCP

Generative AI usage refers to the extent, duration, and frequency of employing generative AI tools (Ghasemaghaei, 2020). Moreover, according to Cheng et al. (2022), SSCP involves three aspects: economic performance (i.e., the overall health and productivity of a firm's supply chain), social performance (i.e., the effect of a firm's supply chain on society and communities), and environmental performance (i.e., the effect of a firm's supply chain on the environment). Hence, we discuss how generative AI usage affects different aspects of SSCP.

Regarding the economic performance of supply chains, numerous manufacturing firms show that generative AI usage can accelerate the innovation process, reduce time to market, and create more efficient and cost-effective products by generating new designs, prototypes, and variations (International Data Corporation, 2024). Further, firms can use generative AI to analyze images, sensor data, and other input types to detect defects and anomalies in products and processes, improving overall product quality and reducing scrap, rework, and warranty costs (Jackson et al., 2024). Finally, firms can use generative AI to produce personalized and customized products, services, and experiences tailored to individual customer preferences and requirements (Kshetri et al., 2023), which can further enhance customer loyalty, increase sales, and drive revenue growth (Li et al., 2023).

For the social performance of supply chains, generative AI usage can automate repetitive or hazardous tasks that pose risks to employee health and safety (Deloitte, 2023). Firms can reduce the likelihood of workplace injuries and occupational hazards by delegating these tasks to generative AI-powered systems. Moreover, generative AI usage can create realistic simulations of workplace environments and potential hazards (Deloitte, 2023). This allows employees to undergo safety training in virtual environments and practice responding to emergencies without real-world risks. Finally, generative AI usage can monitor workplace environments for safety compliance and notify relevant personnel of safety protocol breaches, providing real-time alerts and feedback (IBM, 2024).

Concerning the environmental performance of supply chains, the use of generative AI can allow firms to analyze data on resource usage, energy consumption, and waste generation across the entire supply chain. More importantly, by identifying inefficiencies and areas for improvement, firms can optimize resource allocation, decrease waste, and minimize environmental repercussions (Abou-Foul et al., 2023). Additionally, by integrating environmental criteria into procurement decisions, firms can use generative AI to analyze supplier practices, materials sourcing, and product life cycle data to identify more sustainable options and endorse responsible sourcing practices (Deloitte, 2023). Lastly, by analyzing data on emissions sources, energy usage, and operational activities, firms can use generative AI to identify opportunities for reducing emissions, establish targets for achieving carbon neutrality, and monitor progress over time (Xi et al., 2024). From the arguments presented above, we propose:

H1: Generative AI usage is positively linked with SSCP.

3.3. The mediating role of GSCC

Although generative AI usage theoretically supports various aspects of sustainable SCM (Fosso Wamba et al., 2023), not all firms can effectively utilize it to achieve the expected benefits (IBM, 2024). This gap motivates us to explore further the factors influencing the association between generative AI usage and SSCP. According to the PBV, performance differences among firms stem from their varying approaches to identical practices (Bromiley and Rau, 2014). Hence, we discuss how the two sustainable supply chain practices of GSCC and CEI influence the generative AI usage–SSCP association.

GSCC describes the strategic cooperation between a firm and its supply chain members to execute environmentally friendly activities (Luo et al., 2014). Generative AI has two primary functions: content generation and data augmentation. To achieve GSCC, collaborative decision-making and planning are essential (Ramanathan and Gunasekaran, 2014). Therefore, we propose that generative AI usage influences GSCC primarily through collaborative decision-making and planning. Specifically, generative AI usage facilitates collaborative decision-making processes among supply chain members by providing real-time insights and recommendations (Deloitte, 2023). For example, generative AI-powered decision support systems can help firms evaluate the environmental repercussions of different sourcing options or production methods and identify the most sustainable alternatives. Moreover, generative AI usage enables supply chain members to conduct scenario modeling and simulation exercises to evaluate different planning options and their potential outcomes (Deloitte, 2023). By generating and analyzing multiple scenarios, members can collaboratively identify the optimal plans that balance cost, efficiency, and sustainability considerations.

Regarding the association between GSCC and SSCP, studies have found that GSCC allows supply chain members to share information about, for example, raw materials and energy consumption, laying a solid foundation for optimizing and using resources within

the supply chains (Kong et al., 2021). Further, GSCC facilitates sharing expertise among supply chain members, thereby fostering the development of innovative solutions for sustainability challenges (Hong et al., 2019). Lastly, cultivating collaborative relationships by prioritizing environmental stewardship and social responsibility can establish a more stable and resilient supply chain, further improving a firm's ability to withstand disruptions and turmoil (Scholten and Schilder, 2015). In short, although generative AI usage can potentially improve SSCP, firms can only realize this potential through effective integration with GSCC. This is because all technologies, including generative AI, require specific operational application scenarios to be implemented. Hence, we propose:

H2: GSCC mediates the generative AI usage–SSCP association.

3.4. The mediating role of CEI

Although GSCC facilitates the sharing of information and knowledge among supply chain members (Kong et al., 2021), it is imperative to recognize that simply acquiring new information and knowledge is insufficient because SSCP depends on transforming and upgrading existing supply chain processes. For this reason, we consider another supply chain practice, CEI, which elucidates how firms can efficiently perform value-adding activities by adopting circular strategies (Kristoffersen et al., 2021).

Following Kristoffersen et al. (2021), we conceptualize CEI as a second-order construct with three subdimensions: (a) reinvent and rethink, (b) restore, reduce, and avoid, and (c) recirculate. Specifically, the first subdimension emphasizes developing modular, repairable products and services; the second subdimension aims to increase the recycling potential in production by minimizing material and process waste; and the third subdimension involves repurposing, refurbishing, and remanufacturing products to prolong their life cycle. Therefore, we discuss generative AI usage's influence on the various subdimensions of CEI.

Recall that generative AI offers novel solutions to various problems and unlocks new possibilities in creativity and decision-making (Kshetri et al., 2023). Given these two advantages, firms can use generative AI in the design process to generate numerous iterations of a product or service according to specified parameters and constraints (Deloitte, 2023) and create modular designs that are easier to assemble, disassemble, and repair. Moreover, automation is a significant characteristic of generative AI (IBM, 2024). Such automation allows firms to select materials that are easier to recycle or have a lower environmental effect, minimizing waste and reducing their reliance on non-recyclable materials (Liu et al., 2022). Finally, by analyzing product data and condition assessments, firms can use generative AI to customize refurbishment strategies, extend product life cycles, and minimize downtime (Deloitte, 2023).

Regarding the association between CEI and SSCP, studies have found that CEI reduces the strain on natural resources and minimizes waste generation, leading to more efficient resource utilization across the supply chain (Khan et al., 2021). Moreover, CEI often leads to long-term cost savings because utilizing recycled materials can reduce the need for expensive virgin resources (Cheng et al., 2022). Finally, CEI promotes supply chain resilience by fostering a closed-loop system in which resources are continuously reused and recycled (Kennedy and Linnenluecke, 2022). This reduces reliance on external sources and minimizes exposure to external shocks such as resource scarcity or supply chain disruptions. Overall, considering the potential relationships between generative AI usage and CEI, CEI and SSCP, and embedding generative AI usage in sustainable supply chain practices, we propose:

H3: CEI mediates the generative AI usage–SSCP association.

3.5. The serial mediation role of GSCC and CEI

Sections 3.3 and 3.4 explore the mediating roles of GSCC and CEI in the association between generative AI usage and SSCP. This section explains their relationship by recalling the potential relevance of GSCC and CEI. Given that the potential benefits of GSCC relate mainly to knowledge and information sharing (Hong et al., 2019) and CEI consists of three distinct dimensions (Kristoffersen et al., 2021), we discuss how GSCC influences the different dimensions of CEI from the perspective of knowledge and information sharing.

From a knowledge-sharing perspective, GSCC enables supply chain members to share best practices for designing products with modular components that are easier to disassemble and repair, helping them to adopt and incorporate more sustainable design principles into their products (Ayala et al., 2017). In addition, by sharing knowledge on efficient recycling processes and technologies, supply chain members can optimize the use of raw materials and reduce the amount of waste generated during production (Hicks, 2007). Finally, GSCC facilitates exchanging knowledge and best practices on reuse, refurbishment, and remanufacturing processes. In theory, this exchange enables supply chain members to innovate techniques and materials, extending product life cycles while maintaining quality and performance standards (Vendrell-Herrero et al., 2022).

From an information-sharing perspective, GSCC promotes the standardization of modular components and interfaces among products and suppliers, enhancing the compatibility and interchangeability of parts and simplifying repair and maintenance procedures (Zhou and Benton, 2007). Further, GSCC facilitates the exchange of information among supply chain participants regarding sustainable and recyclable materials. This promotes a deeper understanding of materials' characteristics, availability, and recyclability. Consequently, supply chain members can make well-informed choices when selecting materials and reduce waste generation during production (Scruggs et al., 2016). Ultimately, GSCC facilitates the exchange of regulatory requirements across supply chain participants, enabling them to guarantee that repurposing, refurbishing, and remanufacturing operations comply with legal norms and mitigate regulatory liabilities (Shibin et al., 2020). Thus, given the potential link between GSCC and CEI and their mediating effects on the association between generative AI usage and SSCP, we propose:

H4: GSCC and CEI serially mediate the generative AI usage–SSCP association.

4. Methods

4.1. Data collection

We chose Chinese manufacturing firms for several reasons. First, the Chinese government has actively supported adopting AI and sustainable industry practices (Deloitte, 2023). This supportive environment makes China an ideal setting to examine how generative AI usage promotes supply chain sustainability. Second, unlike information technology firms serving primarily as technology providers, generative AI can be applied across diverse manufacturing domains such as predictive maintenance, production planning, inventory management, and logistics optimization (Fosso Wamba et al., 2023). Thus, investigating manufacturing firms may yield more substantial insights. Third, Chinese manufacturing firms have increasingly developed and adopted generative AI. For example, Haier, a leading home appliance manufacturer, has launched COSMO-GPT, which covers functions such as intelligent question and answer, content generation, control code generation, database querying, decision support, and operations planning (COSMOplat, 2024). Another example is Baosteel leveraging Huawei's PanguLM to achieve precise prediction and intelligent optimization of raw materials, leading to an annual revenue increase of more than 90 million RMB (Huawei, 2024). These examples indicate that Chinese manufacturing firms may provide a rich data source for our analysis.

We initially used a back-translation technique to verify the accuracy and efficacy of the final Chinese questionnaire. This approach involved an expert translator converting the initial English survey into Chinese and another proficient translator interpreting the previously translated Chinese questionnaire into English. Next, we compared the original English items and the back-translated version and made appropriate modifications to correct discrepancies. Following this, we solicited feedback from technology and operations management experts to refine the questionnaire further. Finally, we partnered with a professional survey organization to use their sampling pool for the pre-test and distribution of the questionnaire.

Specific criteria were used to identify suitable respondents. First, respondent firms were required to have adopted AI, as evidenced by their investments in this area over the past three years. Second, respondents were required to be in senior management positions, determined by asking for their job titles. Third, respondents were expected to have expertise in generative AI, which was assessed according to their understanding of its definition and practical applications. Of the 624 firms contacted, 335 responded within the specified time frame, yielding a response rate of 53.69 % for this research. Finally, because our study focuses on manufacturing firms, we retained responses only from these firms, resulting in 213 valid responses. Table 1 details the demographic characteristics of the responding firms.

4.2. Measures

The present study used four primary constructs. First, three items derived from Ghasemaghaei's (2020) research were employed to evaluate generative AI usage. In addition, following Luo et al.'s (2014) study, GSCC was assessed using five specific items. CEI and SSCP were two second-order constructs in our research. A set of nine items, adapted from Kristoffersen et al. (2021), was used to measure three subdimensions of CEI, namely (a) reinvent and rethink; (b) restore, reduce, and avoid; and (c) recirculate. SSCP embodies three distinct aspects: economic, social, and environmental performance. We used 13 items from research conducted by Cheng et al. (2022) to quantify these aspects. Fig. A1 (see Appendix) provides the details for each item, which were aligned on a 7-point Likert scale: descriptors ranged from "1 = firmly disagree (or significantly inferior)" to "7 = firmly concur (or considerably superior)." Finally, we controlled the age and size of the respondent firms.

4.3. Nonresponse bias and common method bias

We implemented several strategies to mitigate nonresponse bias—including offering incentives, sending reminders, and using multiple modes of contact—to ensure the survey was short and easy to complete. We also continuously monitored response rates,

Table 1
Demographic profile.

Firm information	Frequency	Percentage
Years established		
<6 years	2	0.94
6–10 years	28	13.15
11–15 years	52	24.41
16–20 years	82	38.50
>20 years	49	23.00
Number of employees		
≤299	51	23.94
300–999	110	51.64
≥1000	52	24.41
Ownership		
State-owned	42	19.72
Privately owned	151	70.89
Others	20	9.39

allowing us to make real-time adjustments. Finally, we compared the characteristics of the responding firms with those of the non-responding firms to identify any significant differences. Given the similarity in firm age and size between these two groups ($p > 0.05$), it is unlikely that nonresponse bias significantly affects this study (Armstrong and Overton, 1997).

In terms of the common method bias, we first assured respondents of the anonymity and confidentiality of their responses. This measure might encourage respondents to provide open, accurate information without fear of judgment or repercussions. In addition, we randomized the order of questions or items within the survey or questionnaire to reduce response bias resulting from question order effects. Finally, we incorporated statistical techniques into our data analysis process, namely Harman's single factor test. The results indicate that the model fit was unsatisfactory when all items were loaded into a single factor, with $\chi^2 = 1,677.933$, $df = 405$, $\chi^2/df = 4.143$, incremental fit index (IFI) = 0.598, Tucker–Lewis Index (TLI) = 0.565, comparative fit index (CFI) = 0.595, and root-mean-square error of approximation (RMSEA) = 0.122. Overall, common method bias should not significantly affect our study.

5. Results

5.1. Reliability and validity

Internal consistency reliability assesses the extent to which items within a test consistently measure the same construct. To evaluate this, we first used Cronbach's alpha coefficients. Fig. A1 (see Appendix) shows that all Cronbach's alpha coefficients exceeded the threshold of 0.7, thus satisfying the rule of thumb (Nunnally, 1978). We then conducted the confirmatory factor analysis and found the model fit was satisfactory, with $\chi^2 = 561.567$, $df = 377$, $\chi^2/df = 1.490$, IFI=0.942, TLI=0.932, CFI=0.941, and RMSEA=0.048. These results indicated that the hypothesized model was well-aligned with the observed data. The confirmatory factor analysis also provided the standardized factor loadings for each item. Fig. A1 (see Appendix) shows that the standardized factor loadings ranged from 0.648 to 0.886. Further, the composite reliability and average variance extracted (AVE) from each construct exceeded the thresholds of 0.7 and 0.5, respectively, which again confirmed internal consistency reliability and construct reliability (Bentler, 1990). To assess whether a construct has appropriate correlations with other theoretically distinct constructs, we conducted a discriminant validity test by comparing the square roots of the AVEs with pairwise correlations (Fornell and Larcker, 1981). Table 2 shows that all constructs passed the above tests.

5.2. Hypotheses testing

AMOS 21.0 was used to test our hypotheses because of its user-friendly graphical interface, which allows researchers to analyze complex relationships between observed and latent variables. In addition, both direct and indirect effects can be examined within a single model, providing a comprehensive understanding of the phenomena of interest (Kuzma and Sehnem, 2023). Before formally testing the hypotheses, we conducted a collinearity test. We found that the largest variance inflation factor was 1.966, below the commonly accepted threshold of 5 (Hair Jr et al., 2013). Finally, as we introduced earlier, we manipulated generative AI usage and GSCC as unidimensional constructs and CEI and SSCP as second-order constructs.

The results of the structural model are presented in Fig. 2. The model fit indices are satisfactory: $\chi^2 = 687.800$, $df = 449$, $\chi^2/df = 1.532$, IFI=0.926, TLI=0.917, CFI=0.925, and RMSEA=0.050. From the perspective of direct effects, although generative AI usage does not pose a significant link with SSCP ($\beta = 0.129$, $p = 0.17$), we still observe its positive associations with GSCC ($\beta = 0.508$, $p < 0.001$) and CEI ($\beta = 0.453$, $p < 0.001$). There is also a significant association between GSCC and CEI ($\beta = 0.236$, $p < 0.001$). Finally, GSCC ($\beta = 0.278$, $p < 0.001$) and CEI ($\beta = 0.492$, $p < 0.001$) are two factors that can directly influence SSCP.

To further confirm the mediating role of GSCC and CEI in the association between generative AI usage and SSCP, we used the bootstrap approach with 5,000 bootstrap samples. Table 3 illustrates that all the lower boundaries of the 95 % confidence interval (CI) are above zero, indicating the significance of all three indirect effects. Finally, we recalled that the direct effect of generative AI usage on SSCP was insignificant when we simultaneously included generative AI usage, GSCC, and CEI in the estimation. To this end, we removed GSCC and CEI and re-estimated the structural model. In this case, generative AI usage has a significant relationship with SSCP ($\beta = 0.572$, $p < 0.001$). From the above results, we can conclude that GSCC and CEI mediate the association between generative AI usage and SSCP, confirming all the hypotheses of this paper.

Table 2

Mean, standard deviation, and correlations.

	Mean	Standard deviation	1	2	3	4	5	6	7	8
1. Generative AI usage	5.620	0.727	0.743							
2. GSCC	5.919	0.717	0.295**	0.737						
3. Reinvent and rethink	5.315	1.083	0.375**	0.164*	0.838					
4. Restore, reduce, and avoid	5.891	0.632	0.401**	0.336**	0.380**	0.717				
5. Recirculate	5.753	0.656	0.504**	0.390**	0.447**	0.514**	0.773			
6. Economic performance	5.910	0.658	0.481**	0.426**	0.328**	0.389**	0.349**	0.746		
7. Social performance	5.964	0.697	0.377**	0.370**	0.372**	0.381**	0.584**	0.373**	0.756	
8. Environmental performance	5.979	0.670	0.382**	0.383**	0.341**	0.409**	0.433**	0.637**	0.437**	0.739

Notes: ** and * indicate p-values < 0.01 and 0.05, respectively; the square roots of AVEs are in the diagonal.

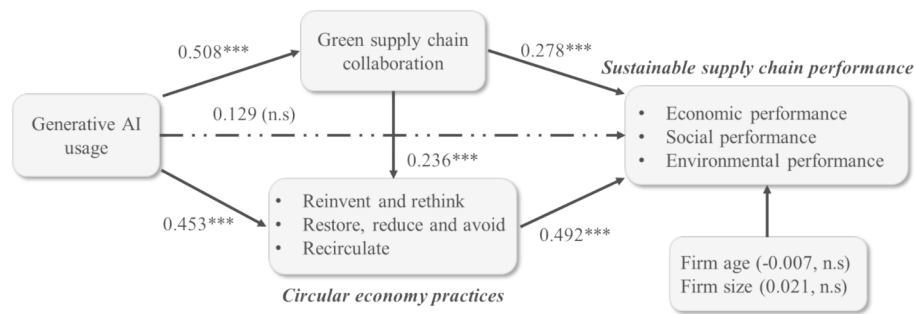


Fig. 2. Structural model results.

Table 3

Indirect effect of generative AI usage on SSCP.

Paths	Estimate	Lower	Upper	P-values
Generative AI usage → GSCC → SSCP	0.141	0.020	0.363	0.033
Generative AI usage → CEI → SSCP	0.223	0.038	0.419	0.037
Generative AI usage → GSCC → CEI → SSCP	0.059	0.011	0.134	0.039
TOTAL	0.423	0.234	0.658	0.002

Note: GSCC=green supply chain collaboration, CEI=circular economy implementation, and SSCP=sustainable supply chain performance.

5.3. Robustness analysis

We checked the robustness of our findings from three aspects. First, motivated by Liang et al.'s (2017) work, we validated our model using another statistical technique. Specifically, we tested the proposed model using the PROCESS macro in SPSS 23.0. For this purpose, we first averaged all variables. Then, using Model 6 in the PROCESS macro, we found that generative AI usage poses a positive effect on SSCP ($\beta = 0.164$, $p < 0.001$). Moreover, such a positive association is mediated by three paths: the mediating effect of GSCC (indirect effect = 0.065, 95 % CI [0.014, 0.185]), the mediating effect of CEI (indirect effect = 0.134, 95 % CI [0.053, 0.230]), and the serial mediating effect of GSCC and CEI (indirect effect = 0.019, 95 % CI [0.003, 0.050]).

Second, we randomly selected subsamples comprising one-half and three-quarters of the full sample ($n_1 = 106$; $n_2 = 159$). This random selection allows us to test whether the observed relationships remain consistent across different subsamples, thus minimizing the risk of our results being influenced by any particular subset of data. We conducted analyses separately for each subsample using the PROCESS macro. The results were consistent across both subsamples, with generative AI usage maintaining a positive effect on SSCP. Moreover, the mediating effect of GSCC (subsample 1: indirect effect = 0.055, 95 % CI [0.009, 0.181]; subsample 2: indirect effect = 0.133, 95 % CI [0.041, 0.244]), the mediating effect of CEI (subsample 1: indirect effect = 0.150, 95 % CI [0.030, 0.275]; subsample 2: indirect effect = 0.086, 95 % CI [0.028, 0.176]), and the serial mediating effect of GSCC and CEI (subsample 1: indirect effect = 0.033, 95 % CI [0.005, 0.097]; subsample 2: indirect effect = 0.032, 95 % CI [0.007, 0.075]), showed similar patterns and significance levels to those identified in the full sample analysis.

Third, because investment in AI technologies often reflects a firm's strategic decision to prioritize and integrate these technologies into its operations, we used AI investment as a proxy for generative AI usage. AI investment was measured by a single item: "How much has your company spent on AI in the past three years? (<1 million RMB; 1 million to 4.99 million RMB; 5 million to 9.99 million RMB; 10 million to 14.99 million RMB; 15 million to 19.99 million RMB; 20 million to 24.99 million RMB; > 25 million RMB)." Using the PROCESS macro, we found that AI investment is positively related to SSCP ($\beta = 0.095$, $p < 0.01$). Notably, AI investment enhances SSCP through three indirect paths: the mediating effect of GSCC (indirect effect = 0.040, 95 % CI [0.009, 0.107]), the mediating effect of CEI (indirect effect = 0.060, 95 % CI [0.018, 0.107]), and the serial mediating effect of GSCC and CEI (indirect effect = 0.018, 95 % CI [0.006, 0.040]). All the above results are summarized in Table A1 (see Appendix). Overall, these results collectively provide strong evidence that our findings are consistent and reliable across different model specifications, sample splits, and alternative measures of generative AI usage.

6. Discussion

This section provides a comparative analysis and discussion of our research findings and existing literature. First, prior research has focused predominantly on discriminative AI and affirmed its positive influence on performance (Benzidia et al., 2021; Enholm et al., 2022; Belhadi et al., 2024). Aligning with previous research, we also confirm the positive effect of generative AI usage on SSCP. This can be attributed to generative and discriminative AI's similar roles in improving decision quality and efficiency (Collins et al., 2021; Jackson et al., 2024). More specifically, both types of AI can integrate multiple variables and provide comprehensive information to

help managers make more accurate and comprehensive decisions and improve overall performance.

Further, despite previous studies exploring AI's effect on supply chain resilience and supply chain finance (Olan et al., 2022a; Dey et al., 2023), we emphasize the role of generative AI in promoting supply chain sustainability. In contrast to research examining the individual roles of GSCC and CEI in enhancing sustainable performance (Luo et al., 2014; Marrucci et al., 2022), this paper highlights the full mediating role of GSCC and CEI in the association between generative AI usage and SSCP. This finding suggests that although generative AI offers a transformative opportunity for supply chains to optimize processes, improve decision-making, and reduce environmental repercussions (Jackson et al., 2024), its potential benefits can only be fully realized when coupled with a commitment to GSCC and circular economy principles.

Finally, our findings align with the PBV. Specifically, the PBV highlights that simply possessing resources or technologies will not lead to superior performance; rather, organizations need to develop and implement particular practices and routines (Bromiley and Rau, 2014). This paper demonstrates that generative AI usage's influence on SSCP is mediated by how firms practice GSCC and CEI. This means firms must go beyond merely adopting generative AI; they need to embed generative AI into their supply chain practices and align them with sustainability goals. The successful implementation of generative AI requires a holistic approach that incorporates technology, processes, and sustainability practices. Thus, our finding reinforces the PBV assertion that practices are crucial for achieving superior performance (Dubey et al., 2022).

6.1. Theoretical implications

We contribute to the existing SCM literature in the following ways. First, while previous studies have extensively discussed the roles of various technologies, particularly discriminative AI (Benzidia et al., 2021; Frank, 2021), in enhancing sustainable SCM, they primarily focus on how these technologies improve operational efficiency, increase transparency, and facilitate better decision-making (Santharm and Ramanathan, 2022). Unlike discriminative AI, which enhances existing operations, generative AI offers a fundamentally different approach by enabling the creation of novel solutions that were previously unattainable, thereby introducing new possibilities for innovation in supply chain processes (Fosso Wamba et al., 2023). Our study explores how to effectively integrate generative AI into SCM practices, shifting the focus from discriminative AI to the untapped potential of generative AI and fostering a deeper understanding of its transformative influence on the future of sustainable SCM.

Second, our study represents a pioneering effort to elucidate the role of generative AI in sustainable SCM (Fosso Wamba et al., 2023). However, unlike Richey et al. (2023) and Jackson et al. (2024), who aim to provide a foundational understanding of how generative AI could revolutionize operations and SCM, we draw on the PBV and provide robust empirical evidence on generative AI usage's effect on SSCP. This empirical approach allows us to move beyond speculative discussions and provide concrete evidence of how to strategically leverage generative AI to improve supply chain sustainability. Moreover, we substantiate the PBV's theoretical proposition that the value of technological resources is contingent upon their effective integration into organizational practices (Bromiley and Rau, 2014).

Third, previous studies have argued that GSCC, which involves collaboration among supply chain partners to achieve environmental goals, and CEI, which focuses on reducing, reusing, and recycling resources, are crucial for driving sustainability in supply chains (Luo et al., 2014; Marrucci et al., 2022). However, in adapting to the new norms of the digital age, it is essential to further explore how these practices interact with emerging technologies, particularly generative AI, to influence supply chain sustainability (Meier et al., 2023). Our study affirms the importance of GSCC and CEI and elucidates how these practices can serve as critical pathways for generative AI usage to enhance SSCP. Hence, our findings advance existing knowledge on the role of GSCC and CEI in sustainable SCM and enrich the current literature by demonstrating how firms can effectively manage and utilize generative AI to bolster their supply chain sustainability efforts.

6.2. Managerial implications

Our findings reveal some managerial implications. First, we underscore the role of GSCC as a mediator in the generative AI usage–SSCP association. This finding suggests that supply chain managers can achieve operational excellence and demonstrate their commitment to sustainability by prioritizing GSCC as a conduit for leveraging generative AI. Because generative AI can analyze historical data and market trends to predict demand more accurately (Kshetri et al., 2023), supply chain managers could share these forecasts with supply chain partners, reducing waste and environmental repercussions by aligning supply with actual demand. Moreover, supply chain managers could use generative AI to simulate and model scenarios for sustainable innovation (Deloitte, 2023). This includes creating new product designs, packaging solutions, and logistics optimizations with supply chain partners to reduce environmental repercussions and improve efficiency.

Second, we reveal that CEI is another mediator in the generative AI usage–SSCP association. This finding suggests that supply chain managers should focus on integrating circular economy principles into their AI strategies. Because generative AI can analyze data on product usage, life cycle, and materials and create multiple design iterations (Deloitte, 2023), supply chain managers could leverage generative AI to identify designs that minimize waste and environmental repercussions while enhancing product longevity. Moreover, CEI aims to build a closed-loop system (Zhang et al., 2021). Given that generative AI can optimize reverse logistics processes by identifying optimal routes for product refurbishment, repair, or recycling (IBM, 2024), supply chain managers could utilize generative AI to promote efficient recycling and repurposing of components or materials back into the supply chain.

Third, in addition to finding that GSCC and CEI are critical mediators in the generative AI usage–SSCP association individually, we identify the potential link between them. This correlation implies that supply chain managers must strategically integrate both

practices to achieve long-term sustainability. To this end, supply chain managers could organize regular workshops or training sessions with key supply chain members to discuss sustainable practices and circular economy principles. For instance, a clothing retailer could collaborate with textile suppliers to explore opportunities for using recycled fibers or implementing water-saving dyeing processes. Moreover, they could partner with suppliers to implement a product take-back program that facilitates returning and recycling end-of-life products. For example, a consumer electronics manufacturer could work with component suppliers to establish a system for collecting and refurbishing electronic devices at the end of their lifespan.

Fourth, policymakers also play a crucial role in facilitating the effective integration of generative AI into sustainable supply chain practices. Policymakers could offer financial incentives such as tax credits, grants, and subsidies to firms adopting generative AI that contribute to sustainable supply chain practices. Moreover, policymakers could establish platforms for firms, research institutions, and policymakers to exchange insights and experiences, promoting knowledge sharing, best practices, and collaborative initiatives in generative AI-enabled sustainable SCM. Finally, policymakers should set clear ethical guidelines and standards for developing and deploying generative AI in supply chains. These could include addressing bias, transparency, accountability, and fairness in AI-generated decisions and practices.

6.3. Limitations and future research

Our study has several limitations. First, we derived our findings from structured survey questions, which limits the depth of understanding. Given that integrating generative AI into SCM frameworks is developing, future research should consider enriching our findings by incorporating qualitative methods to explore more nuanced facets. Second, despite our discussion of the prevalent common method biases, we recognize that eliminating such biases is implausible. Therefore, future research should consider collecting data from various respondents within the same organization or merging primary data with secondary sources. Third, our sampling method did not include firms with no AI usage, resulting in the absence of a control group in our analysis. This limitation may affect the estimation results, and future research should consider including non-AI-using firms to provide a more comprehensive comparison. Finally, we derived our observations predominantly from a distinct Chinese environment. To confirm the validity of our findings, future research should test them on a broader variety of cultural contexts, considering the potential variations in cultural orientations and decision-making approaches between Chinese and Western managerial groups.

7. Conclusion

Generative AI offers promising solutions to sustainable SCM challenges; however, not all firms can take advantage of its benefits. We address this by proposing a novel research framework according to the PBV. By analyzing data from 213 Chinese manufacturing firms, we reveal a positive relationship between generative AI usage and SSCP. Further, GSCC and CEI sequentially mediate this association. Our study contributes to the body of knowledge in AI-enabled SCM and provides new insights for firms navigating the dynamic landscape of global supply chains.

CRediT authorship contribution statement

Lixu Li: Conceptualization. **Wenwen Zhu:** Data curation. **Lujie Chen:** Writing – review & editing, Writing – original draft. **Yaoqi Liu:** Methodology, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Generative AI usage (CR = 0.786; AVE = 0.552; Cronbach's α = 0.784)	- Please indicate how often generative AI tools are used in your firm. (0.797) - Please indicate to what extent generative AI tools are used in your firm. (0.664) - On average, please specify the duration to which generative AI tools are used in your firm. (0.762)
Green supply chain collaboration (CR = 0.856; AVE = 0.543; Cronbach's α = 0.855)	<i>To what extent did your firm engage in the following environmental activities with your supply chain members?</i> - Achieve environmental goals collectively. (0.765) - Developing a mutual understanding of responsibilities regarding environmental performance. (0.672) - Working together to reduce environmental impact of your activities. (0.734) - Conducting joint planning to anticipate and resolve environmental-related problems. (0.746) - Making joint decisions about ways to reduce overall environmental impact of your products. (0.764)
Reinvent and rethink (CR = 0.876; AVE = 0.703; Cronbach's α = 0.872)	<i>Please indicate whether you have implemented the listed practices in your firm, or contributed to its implementation for another stakeholder.</i> - We support products during their lifetime through providing spare parts and/or repair services as separate sales offerings. (0.754) - We provide the result or performance of a product as a service instead of selling the physical product. (0.870) - We provide the access or usage of a product as a service instead of selling the physical. (0.886)
Restore, reduce and avoid (CR = 0.760; AVE = 0.514; Cronbach's α = 0.757)	<i>Please indicate whether you have implemented the listed practices in your firm, or contributed to its implementation for another stakeholder.</i> - We source secondary, recycled and/or renewable materials (e.g. industrial symbiosis, using ocean plastics or non-toxic materials). (0.745) - We run a lean and clean production (e.g. use less energy and materials, treat wastes, rework). (0.648) - We optimize product use and operation to extend the product life, minimize energy use, and/or increase product utilization. (0.753)
Recirculate (CR = 0.815; AVE = 0.597; Cronbach's α = 0.812)	<i>Please indicate whether you have implemented the listed practices in your firm, or contributed to its implementation for another stakeholder.</i> - We provide activities for extending the existing use-cycles of products and parts (e.g. upgrade, repair, maintenance). (0.795) - We provide activities for extending products and parts to new use-cycles (e.g. reuse, refurbish, remanufacture). (0.698) - We provide activities for extending the lifespan of materials (e.g. recycle, cascade, energy recovery). (0.819)
Economic performance (CR = 0.862; AVE = 0.557; Cronbach's α = 0.863)	<i>How does your firm's supply chain perform in the following areas compared to key competitors?</i> - Production costs (0.775) - Profits (0.727) - NPD costs (0.785) - Return on investment (0.703) - Inventory holding costs (0.738)
Social performance (CR = 0.798; AVE = 0.571; Cronbach's α = 0.795)	<i>How does your firm's supply chain perform in the following areas compared to key competitors?</i> - Working conditions (0.785) - Workplace safety (0.650) - Employee health (0.822)
Environmental performance (CR = 0.857; AVE = 0.546; Cronbach's α = 0.857)	<i>How does your firm's supply chain perform in the following areas compared to key competitors?</i> - Solid waste (0.717) - Liquid waste (0.718) - Gas emissions (0.706) - Energy consumption (0.749) - Consumption of hazardous and harmful products (0.800)

Note: Numbers in parentheses are factor loadings

Fig. A1. Measurement items

Table A1
Summary of robustness checks.

Paths	PROCESS Macro (full sample)		Subsample 1 (n = 106)		Subsample 2 (n = 159)		AI investment proxy (full sample)	
	Effect	95 % CI	Effect	95 % CI	Effect	95 % CI	Effect	95 % CI
Indirect effect via GSCC	0.065	[0.014, 0.185]	0.055	[0.009, 0.181]	0.133	[0.041, 0.244]	0.040	[0.009, 0.107]
Indirect effect via CEI	0.134	[0.053, 0.230]	0.150	[0.030, 0.275]	0.086	[0.028, 0.176]	0.060	[0.018, 0.107]
Serial indirect effect via GSCC and CEI	0.019	[0.003, 0.050]	0.033	[0.005, 0.097]	0.032	[0.007, 0.075]	0.018	[0.006, 0.040]

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